Practice Final Exam – Simulation Results

ECEn 483/ ME 431

Winter 2026

Name:_______________________________

At the end of the exam, make sure to submit this file as part of your final upload (along with code used to generate these plots)!!

Part I (20pts)	
Part II (20 pts)	
Part III (25 pts)	
Part IV (35pts)	
Total: (100 pts)	

Part 2. Design models

2.2 Insert plot of the output of the simulation model with initial condition $\theta(0) = \theta_e$ and input τ_e directly below this line.

Part 3. PID Control

3.5 Insert a plot that shows both θ and θ^d when θ^d is a square wave with magnitude ± 10 degrees and frequency 0.1 Hz, and when using a PD controller.

3.6 Insert a plot that shows both θ and θ^d when θ^d is a square wave with maginitude ± 10 degrees and frequency 0.1 Hz, and when using a PID controller.

Part 4. Observer based control

4.2. Insert a plot of the step response of the system for the state space controller with an integrator.

4.5. Insert a plot of the step response of the system for the complete observer-based control.

4.6 Insert a plot of the state estimation error.

4.7 Insert a plot of the LQR-based response.